Best-response dynamics of the symmetric 2-player Privacy Externality Game, $\lambda_1 = \lambda_2 = 0.7$
(each column: the mover sweeps its budget; the non-mover's contract $A_{\varepsilon,\delta}$ is flat but its realised A_{PLD} still moves — the externality)

utility $u_i = V - \lambda_i A_{\mathrm{PLD},i}$ (left axis)
non-mover's utility before the step
model value $V(\sigma)$ (shared, right axis)
realised $A_{\mathrm{PLD},i}$ (vulnerability, right axis)
certified $A_{\varepsilon,\delta,i}$ (contract, right axis)
mover's optimal next choice (mover's row)
row player's ε_i before the step
infeasible candidate
Initial state
Step 1: P1 best-responds
Step 2: P2 best-responds
Step 3: P1 best-responds
Step 4: P2 best-responds
Step 5: P1 best-responds
Step 6: P2 best-responds
Step 7: P1 best-responds
Step 8: P2 best-responds
Step 9: P1 best-responds
Step 10: P2 stays (converged)
Step 11: P1 stays (converged)
Player 1
Player 2
u_1
u_2
$V(\sigma)$, advantage
10^0
10^1
P1 candidate ε_1
P2 candidate ε_2
$(\varepsilon_1, \varepsilon_2) = (5.9, 5.9)$
$\sigma = 1.17$
$u_1 = 0.285,\ A_1 = 0.198$
$u_2 = 0.285,\ A_2 = 0.198$
$(\varepsilon_1, \varepsilon_2) = (6.4, 5.9)$
$\sigma = 1.14$
$u_1 = 0.286,\ A_1 = 0.212$
$u_2 = 0.299,\ A_2 = 0.193$
$(\varepsilon_1, \varepsilon_2) = (6.4, 5.4)$
$\sigma = 1.16$
$u_1 = 0.274,\ A_1 = 0.216$
$u_2 = 0.299,\ A_2 = 0.179$
$(\varepsilon_1, \varepsilon_2) = (6.9, 5.4)$
$\sigma = 1.14$
$u_1 = 0.274,\ A_1 = 0.231$
$u_2 = 0.314,\ A_2 = 0.174$
$(\varepsilon_1, \varepsilon_2) = (6.9, 3.6)$
$\sigma = 1.24$
$u_1 = 0.222,\ A_1 = 0.248$
$u_2 = 0.315,\ A_2 = 0.116$
$(\varepsilon_1, \varepsilon_2) = (10, 3.6)$
$\sigma = 1.08$
$u_1 = 0.226,\ A_1 = 0.340$
$u_2 = 0.405,\ A_2 = 0.084$
$(\varepsilon_1, \varepsilon_2) = (10, 1.2)$
$\sigma = 1.18$
$u_1 = 0.163,\ A_1 = 0.363$
$u_2 = 0.415,\ A_2 = 0.003$
$(\varepsilon_1, \varepsilon_2) = (16, 1.2)$
$\sigma = 0.95$
$u_1 = 0.197,\ A_1 = 0.467$
$u_2 = 0.524,\ A_2 = 0.000$
$(\varepsilon_1, \varepsilon_2) = (16, 1.7)$
$\sigma = 0.95$
$u_1 = 0.200,\ A_1 = 0.466$
$u_2 = 0.524,\ A_2 = 0.002$
$(\varepsilon_1, \varepsilon_2) = (22, 1.7)$
$\sigma = 0.81$
$u_1 = 0.215,\ A_1 = 0.558$
$u_2 = 0.605,\ A_2 = 0.000$
$(\varepsilon_1, \varepsilon_2) = (22, 1.7)$
$\sigma = 0.81$
$u_1 = 0.215,\ A_1 = 0.558$
$u_2 = 0.605,\ A_2 = 0.000$
$(\varepsilon_1, \varepsilon_2) = (22, 1.7)$
$\sigma = 0.81$
$u_1 = 0.215,\ A_1 = 0.558$
$u_2 = 0.605,\ A_2 = 0.000$